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MONITORING DEVICES AND ITS INSTALLATION METHODS FOR SEABED SOIL LIQUEFACTION

Abstract

A seabed soil liquefaction monitoring device for installation on a seabed soil to monitor liquefaction status, includes a first monitoring unit exposed above the seabed soil, including a data logger for recording the measurement data of a soil pressure transducer and a piezometer; an accelerometer for measuring the acceleration of the device under the action an earthquake; and a second monitoring unit fixed to the first monitoring unit and extending into the seabed soil, and including a lower cover with a first opening and at least one second opening; a suction bucket to penetrate the seabed soil, including the soil pressure transducer and the piezometer; wherein the first opening passes through an electrical conduit to accommodate signal lines and power lines of the seabed soil liquefaction monitoring device, while the at least one second opening passed through a drainage pipe to drain water from the suction bucket.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application No. 113108794, filed 2024 Mar. 11. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

[0002] This invention relates to a seabed soil liquefaction monitoring device and its installation method, mainly through a workboat to place the monitoring device in the seabed area to monitor the liquefaction status of the seabed soil, as well as the method of transmitting monitoring data.

BACKGROUND

[0003] Currently, there have been no actual cases of monitoring soil liquefaction in marine environment, only examples such as soil liquefaction monitoring devices and installation methods for terrestrial areas. Referring to the soil liquefaction monitoring device and method proposed in Taiwanese Patent 1821140, as shown in FIG. **6**, the terrestrial soil liquefaction monitoring device and method use a mini suction bucket in combination with a piezometer and soil pressure transducer. Through a suction installation method, the equipment is prevented from sinking into or floating up from liquefied soil when soil liquefaction occurs. However, the terrestrial soil liquefaction monitoring device cannot be directly applied to the marine environment, the main reasons for which are: Due to the deep water environment of offshore marine areas being too far from the shore, the monitoring device cannot directly obtain power supply and telecommunication signals like terrestrial soil liquefaction monitoring devices, therefore it needs to be equipped with independent power supply systems and real-time signal transmission systems in order to meet the objective of transmitting measurement data in real-time for remote monitoring; The main cause of terrestrial soils liquefaction is primarily attributed to seismic loads. Since the dense seismic network on terrestrial areas, the seismic and monitoring data from these seismic stations could be cross-referenced and confirmed. However, the characteristics of the marine environment are complex, seabed soil can experience excess pore water pressure triggered by various causes beyond seismic loads alone. The repeated impacts of extreme wave loads on the seabed can also induce this situation. Due to the lack of seismic stations in offshore areas, it is challenging to directly obtain information on whether an earthquake has occurred on the seabed and its magnitude. This makes it difficult to compare and verify the measurement data, and to determine whether the excess pore water pressure originates from the action of extreme wave loads or seismic loads. Consequently, the seabed soil liquefaction monitoring device needs to be specially designed to consider how to distinguish the cause behind the excitation of excess pore water pressure.

[0004] Therefore, this application proposes a structure design and installation method for a soil liquefaction monitoring device in marine environments, solving the problem of the monitoring device in marine environments being unable to directly obtain power supply and telecommunication signals, and further addressing the issue of being unable to compare and analyze measurement data with seismic data in such marine environments.

SUMMARY

[0005] A seabed soil liquefaction monitoring device designed for installation on a seabed soil to monitor its liquefaction status, comprising: a first monitoring unit exposed above the seabed soil, comprising: a data logger for recording the measurement data of a soil pressure transducer and a piezometer; an accelerometer for measuring the acceleration of the seabed soil liquefaction

monitoring device under the action of an earthquake; and a second monitoring unit fixed to the first monitoring unit and extending into the seabed soil, the second monitoring unit comprising: an lower cover with a first opening and at least one second opening; a suction bucket to penetrate the seabed soil, comprising: the soil pressure transducer for measuring soil stress of the seabed soil; and the piezometer for measuring pore water pressure of the seabed soil; wherein the first opening is used for passing through an electrical conduit to accommodate signal lines and power lines of the seabed soil liquefaction monitoring device, while the at least one second opening is used for passing through a drainage pipe to drain water from the suction bucket.
[0006] The seabed soil liquefaction monitoring device according to this invention, wherein the suction bucket further comprises multiple probes that extend into the seabed soil for connecting the accelerometer and transmitting seismic wave data.
[0007] The seabed soil liquefaction monitoring device according to this invention, the first monitoring unit is a sealed chamber used to prevent external seawater from entering.
[0008] The seabed soil liquefaction monitoring device according to this invention, the at least second opening are set to three openings and evenly spaced around the lower top cover, along with the first opening.
[0009] The seabed soil liquefaction monitoring device according to this invention, wherein, the piezometer and soil pressure transducer are installed inside the suction bucket and positioned near the first opening.
[0010] The seabed soil liquefaction monitoring device according to this invention, wherein, the first monitoring unit further comprising a battery, which provides electric power to the data logger, accelerometer, soil pressure transducer, and piezometer. The battery can store enough electricity for at least five days' use while sunless.
[0011] The seabed soil liquefaction monitoring device according to this invention, further comprising a buoy system connecting the electrical conduit and drainage pipe, the buoy system comprising: a signal transmission system, used for transmitting the data measured from the seabed soil liquefaction device for remote monitoring; and a solar photovoltaic system installed on the buoy system, used for generating and transmitting electricity to the battery through the electrical conduit.
[0012] The seabed soil liquefaction monitoring device according to this invention, wherein, the first monitoring unit is a first cylindrical shape, and the bottom surface of the first monitoring unit is smaller than the top surface of the second monitoring unit in order to accommodate the first opening and at least one second opening.
[0013] The seabed soil liquefaction monitoring device according to this invention, wherein, the first opening and at least one second opening are positioned opposite each other on both sides of the first monitoring unit.
[0014] The seabed soil liquefaction monitoring device according to this invention, wherein, the second monitoring unit is a second cylindrical shape, and the circular base area of the second cylindrical shape is larger than the circular base area of the first cylindrical shape of the first monitoring unit, in order to maintain the structural stability of the seabed soil liquefaction monitoring device.
[0015] The seabed soil liquefaction monitoring device according to this invention, wherein, the height of one side of the cylindrical shape of the second monitoring unit is greater than the height of one side of the cylindrical shape of the first monitoring unit, in order to maintain a structural stability of the seabed soil liquefaction monitoring device.
[0016] The present invention discloses a method for installing a seabed soil liquefaction monitoring device, which is used to penetrate the seabed soil liquefaction monitoring device into the seabed soil, the seabed soil liquefaction monitoring device comprising a first monitoring unit and a second monitoring unit, the second monitoring unit comprising a suction bucket formed by a lower top cover and a lower side wall, the installation method comprising: the first monitoring unit is secured

to a crane on a workboat by using a rope; securing one end of an electrical conduit to the workboat, while the other end of the electrical conduit passing through the first opening of the lower top cover and extending into the suction bucket; connecting one end of a drainage pipe to a pumping device on the workboat, while the other end of the drainage pipe passing through at least one second opening on the lower top cover and extending into the suction bucket; opening the valve of the drainage pipe; controlling the crane to lower the seabed soil liquefaction monitoring device down to the surface of the seabed; releasing the tension of the rope to partially penetrate seabed soil liquefaction monitoring device into the seabed soil; and controlling the pumping device, the water within the suction bucket could be drained through the drainage pipe, allowing the suction bucket to penetrate almost completely into the seabed soil.

[0017] The present invention discloses another method for installing a seabed soil liquefaction monitoring device, which is used to penetrate the device into the seabed soil, the seabed soil liquefaction monitoring device comprising a first monitoring unit and a second monitoring unit; the second monitoring unit comprising a lower top cover and a suction bucket, the installation method includes: connecting a drill rod of a workboat to the first monitoring unit using a rod; securing one end of an electrical conduit to the workboat, while the other end of the electrical conduit passing through a first opening on the lower top cover and extending into the suction bucket; connecting one end of a drainage pipe to a pumping device on the workboat, while the other end of the drainage pipe passing through at least one second opening in the lower top cover and extending into the suction bucket; opening the valve of the drainage pipe; controlling the drill rod to lower the seabed soil liquefaction monitoring device down to the surface of the seabed soil; and controlling the drill rod to provide a necessary resistance to penetrate the suction bucket into the seabed soil, allowing the suction bucket to penetrate almost completely into the seabed soil.

[0018] The method for installing the seabed soil liquefaction monitoring device according to this invention, further comprising: when the seabed soil liquefaction monitoring device is almost completely penetrated into the seabed soil, closing the valve of the drainage pipe to maintain the suction pressure inside the suction bucket, then connecting the drainage pipe and the electrical conduit to a buoy system, by positioning the buoy system, the installation location of the monitoring device can still easily be located and identified in the future.

[0019] The method for installing the seabed soil liquefaction monitoring device according to this invention, further comprising: when the seabed soil liquefaction monitoring device is almost completely penetrated into the seabed soil, injecting a highly fluid and fast-setting material into the drainage pipe, then connecting the drainage pipe and the electrical conduit to a buoy system, by positioning the buoy system, the installation location of the monitoring device can still easily be identified and located in the future.

[0020] The method for installing the seabed soil liquefaction monitoring device according to this invention, the buoy system comprising a signal transmission system, which is used for transmitting the data measured by the seabed soil liquefaction monitoring device timely for remote monitoring; and a solar power system, which is installed on the buoy system for generating and transmitting electricity to the battery through the electrical conduit.

[0021] The method for installing the seabed soil liquefaction monitoring device according to this invention, further comprising placing multiple erosion protection works around the seabed soil liquefaction monitoring device to reduce the possibility of local erosion caused by sea currents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. **1** illustrates the seabed soil liquefaction monitoring device of the present invention.

[0023] FIG. **2** illustrates the inner structure of the first monitoring device of the present invention.

[0024] FIG. **3**. illustrates the inner structure of the second monitoring device of the present invention.
[0025] FIG. **4**A to **4**E illustrates a method for installing the seabed soil liquefaction monitoring device according to this invention.
[0026] FIG. **5** illustrates another method for installing the seabed soil liquefaction monitoring device according to this invention.
[0027] FIG. **6** illustrates a prior art schematic diagram for a terrestrial soil liquefaction monitoring device and method.

DETAILED DESCRIPTION

[0028] Due to the unique environment of the marine environment, in addition to seismic loads, the possible causes of excess pore water pressure in the seabed soil can be triggered by the repetitive action of extreme wave loads. Therefore, in order to identify the causes of excess pore water pressure in the seabed soil detected by the monitoring device, please refer to FIG. **1** for the seabed soil liquefaction monitoring device **100** of the present invention, which is used to detect the liquefaction status of the seabed soil **900** where it is installed. The monitoring device **100** includes a first monitoring unit **120** and a second monitoring unit **140**. The first monitoring unit **120** is a sealed cavity formed by the upper cover **124**, the upper side wall **126**, and the lower cover **144**, which can prevent seawater from entering. It includes a simple joint **122** set on the upper cover **124** to connect the ropes or rods used for subsequent placement of this seabed soil liquefaction monitoring device **100**. The second monitoring unit **140** is a suction bucket **147** formed by the lower cover **144** and the lower side wall **146** with a downward opening. The lower cover **144** has a first opening **148** for the passage of electrical conduit, and at least one second opening **149**, for example, three drainage pipes can be connected individually. In the preferred configuration, these three second openings **149** are uniformly arranged around the lower cover **144**, so that the suction bucket can have balanced downward pressure distribution during the process of removing water and penetration into the seabed soil. When the suction bucket **147** is slightly off balance during the penetration process, the pumping rate of the second opening **149** on the higher side can be increased to accelerate the penetration speed of that side, dynamically adjusting the suction bucket **147** to return to a balanced state. The area surrounded by the upper side wall **126** of the first monitoring unit **120** is smaller than the area surrounded by the lower side wall **146** of the second monitoring unit **140**, and the height of the upper side wall **126** of the first monitoring unit **120** is also smaller than the height of the lower side wall **146** of the second monitoring unit **140**. This design can effectively enhance the structural stability of the monitoring device **100** and ensure that the penetration of the monitoring device **100** in subsequent installation steps is less affected by sea current flow and has less angular and positional deviation.
[0029] Please refer to FIG. **2** for the internal structural diagram of the first monitoring unit **120**. It includes an accelerometer **129**, used to measure the acceleration of the seabed soil liquefaction monitoring device **100** under seismic action. A data logger **127** is used to record the measurement signals from the soil pressure transducer **142** and the piezometer **143**. A battery **128** is used to provide power to the accelerometer **129**, data logger **127**, soil pressure transducer **142**, and piezometer **143**, especially in unstable weather conditions at sea. If the subsequent storage of electricity generated by the solar power system is used, it is ideal to have a storage capacity for at least five days' use while sunless. The first monitoring unit **120** is mainly used to detect nearby seismic waves and can determine whether the excitation of pore water pressure in the soil is caused by an earthquake based on the acceleration measured by the accelerometer **129**. The circuits or power lines of the accelerometer **129**, data logger **127**, and battery **128** can enter the second monitoring unit **140** (not shown in the figure) first and then be transmitted or received from the outside through its first opening **148** and the electrical conduit **300** passing through it. However, the airtightness of the first monitoring unit **120** must be maintained in this structure.
[0030] Please refer to FIG. **3** for the internal structural diagram of the second monitoring unit **140**.

It includes a soil pressure transducer **142**, which is installed on the lower surface of the lower cover **144** to measure the soil stress of the seabed soil. A piezometer **143** is installed on the inner surface of the lower side wall **146** to measure the pore water pressure of the seabed soil. The information collected by the above-mentioned equipment can be used to determine whether liquefaction has occurred in the seabed soil. The soil pressure transducer **142** and the piezometer **143** can be placed close to the first opening **148** and at a distance from the second opening **149**. This is mainly because the first opening **148** is primarily used for the passage of the electrical conduit **300**, making it easier to arrange the signal line layout of the soil pressure transducer **142** and the piezometer **143**. When setting up the suction bucket **147** of the second monitoring unit **140**, there may be a gap between the lower cover **144** and the seabed. Therefore, multiple probes **141** are installed inside the second monitoring unit **140**, which are structurally directly connected to the accelerometer **129** of the first monitoring unit **120**. The ends of the multiple probes **141** can be inserted into the seabed soil, allowing the accelerometer **129** to accurately measure seismic data without being affected by the gap.

[0031] Next, please refer to FIGS. **4**A to **4**E for the installation method of the seabed soil monitoring device **100** of the present invention. First, please refer to FIG. **4**A. Conduct a side-scan sonar survey (not shown in the figure) at the selected installation location on the seabed soil **900**. Secure the monitoring device **100** to the crane **410** on the workboat **400** using a rope **412**. Fix one end of the electrical conduit **300** on the workboat **400**, and pass the other end through the first opening **148** of the lower cover **144** and into the suction bucket **147** of the second monitoring unit **140**. Connect one end of the drainage pipe **200** to a pumping device **420** on the workboat **400**, and pass the other end of the drainage pipe **200** through the second opening **149** of the lower cover **144** and into the suction bucket **147** of the second monitoring unit **140**. Then, open the valve **210** of the drainage pipe **200** to allow seawater inside the suction bucket **147** to be discharged during the subsequent process. Control the crane **410** to lower the monitoring device **100** onto the surface of the seabed soil **900**.

[0032] Next, please refer to FIG. **4**B. When the monitoring device **100** sinks and makes contact with the surface of the seabed soil **900**, at this point, control the crane **410** to release the tension of the rope **412**. The monitoring device **100** will continue to penetrate into the seabed soil **900** due to its own weight.

[0033] Then, referring to FIG. **4**C, control the pumping device to start pumping out the water inside the suction bucket **147** to generate suction pressure, allowing the suction bucket **147** to penetrate almost completely into the seabed soil **900**. At this point, the rope **412** can be removed.

[0034] For the subsequent treatment of drainage pipe **200**, two methods can be used. The first method is to close valve **210**, maintaining suction pressure in the drainage pipe **200** and the suction bucket. Then, directly connect the drainage pipe **200** to the subsurface float **630** (refer to FIG. **4**E) and place it naturally in the sea. The risk of this method is that if the drainage pipe **200** is exposed to unpredictable sea conditions(such as currents and waves), or damage in the future, it may cause the suction pressure inside the suction bucket **147** to disappear, resulting in loosening of the monitoring device **100**. However, the relative advantage of this method is that if you want to remove the monitoring device **100** in the future, you only need to perform the reverse operation, which is to pressurize the suction bucket **147** by injecting water through the drainage pipe **200**. This will automatically cause the monitoring device **100** to float up from the seabed, making it convenient for recovery.

[0035] The second method is to inject a highly fluid and rapidly setting material into the drainage pipe **200**, filling the gap between the lower top cover **144** of the suction bucket **147** and the seabed, as well as some sections of the drainage pipe. This blocks the drainage pipe **200** and causes it to fail naturally, maintaining the suction pressure inside the suction bucket **147**. This method can avoid the risk of suction pressure failure in the future due to pipeline damage. However, it is unable to decommission the suction bucket **147** by reverse water injection into the drainage pipe **200**, as described in the first method. In other words, the difficulty of recovering the monitoring device **100** is higher.

[0036] Referencing FIG. **4**D, multiple erosion protection works **500** are placed around the monitoring device **100** to reduce the possibility of local scouring caused by sea current disturbances.

[0037] Continuing to reference FIG. **4**E, the electrical conduit **300** and drainage pipe **200** are fixed to the subsurface float **630**, and then connected to the buoy system **600**. The buoy system **600** is equipped with a solar photovoltaic system **620** to provide the necessary power for the monitoring device **100**. The signal transmission device **640** can transmit data such as seismic wave information, acceleration information, soil pressure information, pore water pressure information, obtained by the monitoring device **100** to the nearby wind turbine **700**. Then, the wind turbine **700** uses fiber optics to transmit the aforementioned data back to terrestrial areas for remote monitoring.

[0038] Next, please refer to FIG. **5** for another method of installing the soil monitoring device **100** in the sea area. This method uses the drill rod **422** on the work boat **400**, which is connected to the simple joint **122** of the monitoring device **100** through a custom-made drill rod connecting piece **430**. Then, one end of the electrical conduit **300** is fixed on the work boat **400**, and the other end of the electrical conduit **300** passes through the first opening **148** of the lower top cover **144** to extend into the suction bucket **147** of the monitoring device **100**. Next, one end of the drainage pipe **200** is connected to the pumping device **420** on the work boat **400**, and the other end of the drainage pipe **200** passes through the second opening **149** of the lower top cover **144** to extend into the suction bucket **147** of the monitoring device **100**. Then, the valve **210** of the drainage pipe **200** is opened to gradually drain the seawater from the suction bucket **147**. At the same time, the drill rod **422** is controlled to lower the seabed soil liquefaction monitoring device **100** down to the surface of the seabed soil **900**. Then, the drill rod **422** is controlled to provide the necessary resistance for penetrating into the seabed soil **900**, so that the suction bucket **147** is almost fully penetrated into the seabed soil **900**. The subsequent steps of removing the drainage pipe, placing multiple erosion protection works, and connecting the electrical conduit and drainage pipe to the buoy system are similar to those shown in FIGS. **4**D and **4**E, and will not be further elaborated.

[0039] Although the technical content of the present invention has been disclosed in the above paragraph, it is not intended to limit the present invention. Any modifications and improvements made by those skilled in the art without departing from the spirit of the present invention should be encompassed within the scope of the present invention. Therefore, the protection scope of this creation should be determined based on the defined scope of the accompanying patent claims.

Claims

1. A seabed soil liquefaction monitoring device for installation on a seabed soil to monitor liquefaction status, the device comprising: a first monitoring unit exposed above the seabed soil, comprising: a data logger for recording measurement data of a soil pressure transducer and a piezometer; an accelerometer for measuring acceleration of the seabed soil liquefaction monitoring device under the action of an earthquake; and a second monitoring unit fixed to the first monitoring unit and extending into the seabed soil, the second monitoring unit comprising: a lower cover with a first opening and at least one second opening; a suction bucket to penetrate the seabed soil, comprising: the soil pressure transducer for measuring soil stress of the seabed soil; and the piezometer for measuring pore water pressure of the seabed soil; wherein the first opening is used for passing through an electrical conduit to accommodate signal lines and power lines of the seabed soil liquefaction monitoring device, while the at least one second opening is used for passing through a drainage pipe to drain water from the suction bucket.

2. The seabed soil liquefaction monitoring device according to claim 1, wherein the suction bucket further includes multiple probes that extend into the seabed soil for connecting to the accelerometer

and transmitting seismic wave information data.
3. The seabed soil liquefaction monitoring device according to claim 1, the first monitoring unit is a sealed chamber used to prevent external seawater from entering.
4. The seabed soil liquefaction monitoring device according to claim 1, the at least second opening are set to three and evenly distributed around the lower top cover with the first opening.
5. The seabed soil liquefaction monitoring device according to claim 1, wherein, the piezometer and soil pressure transducer are installed inside the suction bucket and positioned near the first opening.
6. The seabed soil liquefaction monitoring device according to claim 1, wherein, the first monitoring unit further comprising a battery, which provides electric power to the data logger, accelerometer, soil pressure transducer, and piezometer. The battery can store electricity for at least five days' use.
7. The seabed soil liquefaction monitoring device according to claim 1, further comprising a buoy system connecting the electrical conduit and drainage pipe, the buoy system comprising: a signal transmission system for transmitting the data measured from the seabed soil liquefaction device; and a solar photovoltaic system, installed on the buoy system, for generating and transmitting electricity to the battery through the electrical conduit.
8. The seabed soil liquefaction monitoring device according to claim 1, wherein, the first monitoring unit is a first cylindrical shape, and the bottom surface of the first monitoring unit is smaller than the top surface of the second monitoring unit in order to accommodate the first opening and at least one second opening.
9. The seabed soil liquefaction monitoring device according to claim 1, wherein, the first opening and at least one second opening are positioned opposite each other on both sides of the first monitoring unit.
10. The seabed soil liquefaction monitoring device according to claim 8, wherein, the second monitoring unit is a second cylindrical shape, and the circular base area of the second cylindrical shape is larger than the circular base area of the first cylindrical shape of the first monitoring unit.
11. The seabed soil liquefaction monitoring device according to claim 10, wherein, the height of one side of the cylindrical shape of the second monitoring unit is greater than the height of one side of the cylindrical shape of the first monitoring unit.
12. A method for installing a seabed soil liquefaction monitoring device, which is used to penetrate seabed soil liquefaction monitoring device into the seabed soil, the seabed soil liquefaction monitoring device comprising a first monitoring unit and a second monitoring unit, the second monitoring unit comprising a suction bucket formed by a lower top cover and a lower side wall, the installation method comprising: securing the first monitoring unit to a crane on a workboat by using a rope; securing one end of an electrical conduit to the workboat, while the other end of the electrical conduit passing through the first opening of the lower top cover and extending into the suction bucket; connecting one end of a drainage pipe to a pumping device on the workboat, while the other end of the drainage pipe passing through at least one second opening in the lower top cover and extending into the suction bucket; opening the valve of the drainage pipe; controlling the crane to lower the seabed soil liquefaction monitoring device down to the surface of the seabed soil; releasing the tension of the rope to partially penetrate seabed soil liquefaction monitoring device into the seabed soil; and controlling the pumping device to drain the water in the suction bucket through the drainage pipe, allowing the suction bucket to penetrate almost completely into the seabed soil.
13. A method for installing a seabed soil liquefaction monitoring device, which is used to penetrate the device into the seabed soil, the seabed soil liquefaction monitoring device comprising a first monitoring unit and a second monitoring unit; the second monitoring unit comprising a lower top cover and a suction bucket, the installation method comprising: connecting a drill rod of a workboat to the first monitoring unit using a rod; securing one end of an electrical conduit to the

workboat, while the other end of the electrical conduit passing through a first opening on the lower top cover and extending into the suction bucket; connecting one end of a drainage pipe to a pumping device on the workboat, while the other end of the drainage pipe passing through at least one second opening on the lower top cover and extending into the suction bucket; opening the valve of the drainage pipe; controlling the drill rod to lower the seabed soil liquefaction monitoring device down to the surface of the seabed soil; and controlling the drill rod to provide a necessary resistance to penetrate the suction bucket into the seabed soil, allowing the suction bucket to penetrate almost completely into the seabed soil.

14. The method for installing the seabed soil liquefaction monitoring device according to claim 12, further comprising: closing the valve of the drainage pipe to maintain the suction pressure inside the suction bucket when the seabed soil liquefaction monitoring device is almost completely penetrated into the seabed soil; and connecting the drainage pipe and the electrical conduit to a buoy system.

15. The method for installing the seabed soil liquefaction monitoring device according to claim 13, further comprising: closing the valve of the drainage pipe to maintain the suction pressure inside the suction bucket when the seabed soil liquefaction monitoring device is almost completely penetrated into the seabed soil; and connecting the drainage pipe and the electrical conduit to a buoy system.

16. The method for installing the seabed soil liquefaction monitoring device according to claim 12, further comprising: injecting a highly fluid and fast-setting material into the drainage pipe when the seabed soil liquefaction monitoring device is almost completely penetrated into the seabed soil; and connecting the drainage pipe and the electrical conduit to a buoy system.

17. The method for installing the seabed soil liquefaction monitoring device according to claim 13, further comprising: injecting a highly fluid and fast-setting material into the drainage pipe when the seabed soil liquefaction monitoring device is almost completely penetrated into the seabed soil; and connecting the drainage pipe and the electrical conduit to a buoy system.

18. The method for installing the seabed soil liquefaction monitoring device according to claim 14, the buoy system comprising: a signal transmission system, which is used for transmitting the data measured by the seabed soil liquefaction monitoring device timely for remote monitoring; and a solar power system, which is installed on the buoy system for generating and transmitting electricity to the battery through the electrical conduit.

19. The method for installing the seabed soil liquefaction monitoring device according to claim 15, the buoy system comprising: a signal transmission system, which is used for transmitting the data measured by the seabed soil liquefaction monitoring device timely for remote monitoring; and a solar power system, which is installed on the buoy system for generating and transmitting electricity to the battery through the electrical conduit.

20. The method for installing the seabed soil liquefaction monitoring device according to claim 16, the buoy system comprising: a signal transmission system, which is used for transmitting the data measured by the seabed soil liquefaction monitoring device timely for remote monitoring; and a solar power system, which is installed on the buoy system for generating and transmitting electricity to the battery through the electrical conduit.

21. The method for installing the seabed soil liquefaction monitoring device according to claim 17, the buoy system comprising: a signal transmission system, which is used for transmitting the data measured by the seabed soil liquefaction monitoring device timely for remote monitoring; and a solar power system, which is installed on the buoy system for generating and transmitting electricity to the battery through the electrical conduit.

22. The method for installing the seabed soil liquefaction monitoring device according to claim 14, further comprising: placing multiple erosion protection works around the seabed soil liquefaction monitoring device to reduce the possibility of local erosion caused by sea currents.

23. The method for installing the seabed soil liquefaction monitoring device according to claim 15, further comprising: placing multiple erosion protection works around the seabed soil liquefaction monitoring device to reduce the possibility of local erosion caused by sea currents.

24. The method for installing the seabed soil liquefaction monitoring device according to claim 16, further comprising: placing multiple erosion protection works around the seabed soil liquefaction monitoring device to reduce the possibility of local erosion caused by sea currents.

25. The method for installing the seabed soil liquefaction monitoring device according to claim 17, further comprising: placing multiple erosion protection works around the seabed soil liquefaction monitoring device to reduce the possibility of local erosion caused by sea currents.